Hypertension

ORIGINAL ARTICLE

Visit-to-Visit Blood Pressure Variability and Cardiovascular Outcomes in Patients Receiving Intensive Versus Standard Blood Pressure Control: Insights From the STEP Trial

Jiachen Yu,* Qirui Song,* JingJing Bai, Shouling Wu, Peili Bu, YuFeng Li, Jun Cai

BACKGROUND: The clinical prognostic value of visit-to-visit blood pressure (BP) variability (BPV) is debatable, and relative studies among patients receiving BP control to achieve lower BP targets are limited.

METHODS: We analyzed a dataset from the STEP trail (Strategy of Blood Pressure Intervention in the Elderly Hypertensive Patients) to investigate the relationship between visit-to-visit BPV and cardiovascular events in patients with hypertensive aged 60 to 80 years. Visit-to-visit BPV was defined as the coefficient of variation, SD, delta, average real variability, and variability independent of the mean of BP measured at 6-, 9-, 12-, 15-, and 18-month follow-up visits. We computed hazard ratios for the risks associated with a 1-SD increase in BPV indexes in multivariable cox regression models.

RESULTS: Among 7678 patients from the STEP trial, after adjustment for multiple confounders, diastolic BPV indexes were significantly associated with the primary composite end point (hazard ratios ≥1.21; *P*≤0.029) in the standard group, while there was no association between the clinical outcomes and systolic BPV (*P*≥0.091). In the intensive treatment group, either systolic or diastolic BPV was no association with clinical outcomes(*P*≥0.30). Sensitivity analyses using an alternative method to calculate BPV based on 7 BP records generated confirmatory results.

CONCLUSIONS: In older adults with hypertension, visit-to-visit diastolic BPV is an independent predictor of adverse health outcomes in the standard treatment group. However, BPV did not have prognostic value in the intensive treatment group.
***(Hypertension.* 2023;80:1507–1516. DOI: 10.1161/HYPERTENSIONAHA.122.20376.) • Supplement Material**.

REGISTRATION: URL: https://www.clinicaltrials.gov; Unique identifier: NCT03015311.

Key Words: blood pressure variability ■ cardiovascular outcomes ■ hypertension

Visit-to-visit blood pressure (BP) variability (BPV) is the measurement of the variation in BP across outpatient visits, which essentially represents the long-term dynamic systematic BP regulation.[1,2] For decades, research has been conducted to determine whether visit-to-visit BPV is an independent factor affecting many kinds of clinical events and whether it should be regard as a treatment target or an important indicator to monitor. Previous studies have demonstrated that visit-to-visit BPV is associated with several adverse clinical outcomes, including all-cause death, cardiovascular mortality, cardiovascular events, heart failure, and stroke.[3–9] However, there were studies failed to report its prognostic value, in which researchers believed BP level instead of BPV that actually matter.[10–12]

The impressive cardiovascular benefits of intensive BP control to a lower BP level have been proven in patients with hypertension.[13–15] In the STEP trial (Strategy of Blood Pressure Intervention in the Elderly

Correspondence to: Jun Cai, Hypertension Center, FuWai Hospital, No. 167, Beilishi Rd, Xicheng District, Beijing 100037, China. Email caijun7879@126.com
*J. Yu and Q. Song are joint first authors.
Supplemental Material is available at https://www.ahajournals.org/doi/suppl/10.1161/HYPERTENSIONAHA.122.20376.
For Sources of Funding and Disclosures, see page 1515

Hypertension is available at www.ahajournals.org/journal/hyp

NOVELTY AND RELEVANCE

What Is New?

This study is one of few research examined the associations of cardiovascular outcomes with visit-to-visit blood pressure variability (BPV) assessed by 5 indexes (coefficient of variation, SD, delta, average real variability, and variability independent of the mean) in hypertensive elders with 2 different blood control targets respectively.

What Is Relevant?

In the standard treatment group, the primary composite end point was significantly associated with diastolic BPV, while systolic BPV did not have prognostic value for cardiovascular disease.

In the intensive treatment group, neither systolic nor diastolic BPV was associated with cardiovascular events.

Sensitivity analyses using alternative methods to calculate visit-to-visit BPV generate confirmatory results.

Clinical/Pathophysiological Implications?

For elder patients with hypertension with standard antihypertension treatment, visit-to-visit diastolic BPV was associated with cardiovascular complications. Thus, it is critical for these patients to monitor diastolic blood pressure during the follow-up period.

Nonstandard Abbreviations and Acronyms

ACS	acute coronary syndrome
BP	blood pressure
BPV	blood pressure variability
MACE	major adverse cardiac events
STEP	Strategy of Blood Pressure Intervention in the Elderly Hypertensive Patients

Hypertensive Patients), we found that receiving antihypertensive treatment to achieve a systolic BP target of 110 to <130 mm Hg results in a significantly lower incidence of cardiovascular events than reduction to a target of 130 to <150 mm Hg.[13] However, it is unclear whether the visit-to-visit BPV has prognostic value in patients receiving these 2 types of BP control strategies. Herein, we conducted a post hoc analysis of the STEP trial to test the hypothesis that visit-to-visit BPV is associated with a higher risk of cardiovascular events in older adults with hypertension, and to examine the prognostic value of BPV in patients receiving intensive BP control.

METHODS

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Study Design and Participants

This study was a post hoc analysis of the STEP trial.[13,16] The STEP trial was a multicenter randomized controlled trial that compared 2 strategies for BP control (intensive treatment with an SBP target of 110 to <130 mm Hg versus standard treatment with an SBP target of 130 to <150 mm Hg) and evaluated the effect of these strategies on cardiovascular outcomes in 8511 older adults with hypertension treated at 42 clinical centers in China. Briefly, between January 10 and December 31, 2017, participants aged 60 to 80 years with treated and untreated essential hypertension (defined as an SBP of 140–190 mm Hg) were enrolled. Patients with a history of ischemic or hemorrhagic stroke were excluded. All eligible participants were randomly assigned to either the intensive treatment group (SBP target of 110 to <130 mm Hg) or the standard treatment group (SBP target of 130 to <150 mm Hg) with the use of a central computerized randomization program. In this post hoc analysis, 608 patients who were missing once and more office BP records were excluded. Furthermore, 225 patients who experienced any of the clinical outcomes in the first 18 months were excluded from the present analysis to avoid BP fluctuations caused by outcome events. The final number of participants included in the present analysis was 7678, comprising 3875 in the intensive treatment group and 3803 in the standard treatment group. All enrolled patients underwent a detailed baseline evaluation.

BP Measurement and Assessment of Visit-to-Visit BPV

Each participant's BP was measured in clinic by trained staff 3 times with an interval of 1 to 2 minutes using the same validated automatic BP monitoring device (Omron HBP-1100U; Omron Healthcare). The average value of the 3 measurements was recorded as the visit BP. As the BP was not stable during the first 6 months, we collected the BP records at the 6-, 9-, 12-, 15-, and 18-month visits to calculate the BPV to avoid BP fluctuations due to medication adjustments as much as possible. We used the coefficient of variation, SD, delta, average real variability, and variability independent of the mean to assess the BPV.[8,17–19] The coefficient of variation was calculated as the SD divided by the mean. Delta BP was defined as the maximum BP minus the minimum BP. The average real variability was calculated as the average of the absolute differences between consecutive BP measurements. The variability independent of the mean was calculated as the SD divided by the mean to the power x and multiplied by the population mean to the power x. All parameters were evaluated for both the systolic BP and diastolic BP.

Covariates

Trained STEP physicians collected information about sociodemographic characteristics, comorbidity, baseline medication, and antihypertension treatment during follow-up, and laboratory examinations, including the fasting blood glucose, triglyceride, total cholesterol, low-density lipoprotein cholesterol, and creatinine concentrations. The 10-year risk of cardiovascular disease was estimated using the Framingham risk score (FRS).[20] The estimated glomerular filtration rate was calculated using the Modification of Diet in Renal Disease equation.[21] Chronic kidney disease was defined as an estimated glomerular filtration rate of <60 mL·min^{-1}·1.73 m^{-2}.

Clinical Outcomes

The clinical outcomes in this study included primary and secondary outcomes. The primary outcome was a composite of death from cardiovascular causes, acute coronary syndrome (ACS; myocardial infarction and hospitalization for unstable angina), acute decompensated heart failure, coronary revascularization, atrial fibrillation, and stroke.[13] The secondary outcomes were major adverse cardiac events (MACE) and ACS. MACE was defined as a composite of the first occurrence of ACS, acute decompensated heart failure, coronary revascularization, and death from cardiovascular causes. All 3 clinical outcomes were identical to those in the STEP trial.[13,16]

Statistical Analysis

Continuous variables are presented as mean±SD, and categorical variables are presented as n (%). We compared means and proportions by ANOVA and by the χ^2 statistic, respectively. To analyze the associations between the BPV and clinical outcomes, we computed the hazard ratios (HRs) with 95% CIs of the clinical outcomes related to each 1-SD increment in the systolic and diastolic BPV indexes using Cox regression models in the randomization group. In multivariable-adjusted Cox regression analyses, we adjusted for sex, age, body mass index, mean corresponding BP during the follow-up period, current smoking status, dyslipidemia, diabetes, previous cardiovascular disease, baseline chronic kidney disease status, the number of antihypertensive drug class received at the 6-month study visit, use of antihypertensive medications by drug class (diuretics, renin-angiotensin inhibitors, calcium channel blockers, β-blockers), and intake of aspirin, lipid-lowering drugs, and antidiabetic agents. All these variables were selected a priori for inclusion in the regression models. Besides, considering the strong prognostic value of systolic BP level, the Cox regression model of diastolic BPV which adjusted for demographic variables and mean systolic BP level during the follow-up period instead of mean diastolic BP level has also been performed in this analysis. Some of the variables in Table 1 and Table S1 were not included in this model to avoid possible overfitting of the model. To validate the results, we performed a sensitivity analysis using an alternative method to calculate BPV based on 7 BP records measured at the 6-, 9-, 12-, 15-, 18-, 21-, and 24-month follow-up visits. To determine which variables affected the BPV, we also performed multivariable linear regression analysis to identify the factors independently correlated with the BPV.

We used R software (version 4.1.2) to perform all analyses. Statistical significance was set as a 2-sided *P* value of <0.05.

RESULTS

Patient Characteristics

Of the 8511 patients enrolled in the STEP trial, 883 patients were excluded from the present analysis in accordance with the exclusion criteria. A total of 7678 patients (90.2%) were analyzed in the current analysis (mean age, 66.19±4.79 years, 53.5% were male), comprising 3875 in the intensive treatment group and 3803 in the standard treatment group (Figure 1). Table 1 and Table S1 showed the baseline characteristics of the patients by the median distributions of diastolic and systolic BPV, respectively. And as shown in Table S2, there were significant differences between the intensive treatment group and the standard treatment group in the mean systolic BP during the period of BPV ascertainment (128 mm Hg in the intensive treatment group and 136 mm Hg in the standard treatment group), mean diastolic BP during the period of BPV ascertainment (76 mm Hg in the intensive treatment group and 79 mm Hg in the standard treatment group), and antihypertensive medication. In addition, compared with the standard treatment group, the intensive treatment group had higher BPV (Table S2).

Associations Between BPV and Clinical Outcomes

During a median follow-up period of 3.34 years, there were 212 primary outcome events (97 in the intensive treatment group and 115 in the standard treatment group), 176 MACE (81 in the intensive treatment group and 95 in the standard treatment group) and 144 patients experienced ACS (70 in the intensive treatment group and 74 in the standard treatment group).

In the analysis, all indexes of diastolic BPV were associated with the primary outcome in the standard treatment group (adjusted HRs, 1.21–1.29; $P \leq 0.029$; Table 2), while 4 indexes of diastolic BPV were related to the occurrence of MACE and ACS in the standard treatment group (adjusted HRs, 1.21–1.33; $P \leq 0.04$; Table 2). The analysis of the intensive treatment group revealed no significant associations between BPV indexes (both systolic and diastolic BPV) and all adverse health outcomes (including the primary outcome, MACE, and ACS) in both unadjusted ($P \geq 0.14$; Table 3) and multivariable-adjusted models ($P \geq 0.30$; Table 3). The systolic BP-adjusted Cox model also generated consistent results [adjusted HRs, 1.20–1.28; $P \leq 0.041$ for primary end point in the standard group (Table S3), while $P \geq 0.20$ in the intensive group (Table S4)].

Sensitivity Analyses

Sensitivity analyses performed using 7 follow-up BP values to calculate the BPV confirmed the above-mentioned results. For 7590 patients included in the

Table 1. Baseline Characteristics of Participants by Median of Indexes of Diastolic Blood Pressure Variability

Characteristics	CV		SD		Delta		ARV		VIM	
	<0.05	≥0.05	<4.18	≥4.18	<11.30	≥11.30	<4.85	≥4.85	<8.82	≥8.82
N (%)	3846 (50.1)	3832 (49.9)	3845 (50.1)	3833 (49.9)	3823 (49.8)	3855 (50.2)	3836 (50.0)	3842 (50.0)	3837 (50.0)	3841 (50.0)
Age, mean (SD)	66.1 (4.77)	66.3 (4.82)*	66.2 (4.79)	66.2 (4.80)	66.2 (4.79)	66.2 (4.80)	66.2 (4.81)	66.2 (4.78)	66.2 (4.79)	66.2 (4.80)
Sex (male), n (%)	2049 (53.3)	2061 (53.8)	2079 (54.1)	2031 (53.0)	2074 (54.3)	2036 (52.8)	2079 (54.2)	2031 (52.9)	2074 (54.1)	2036 (53.0)
BMI, mean (SD)	25.60 (3.14)	25.57 (3.18)	25.53 (3.14)	25.64 (3.19)	25.54 (3.13)	25.63 (3.19)	25.49 (3.12)	25.68 (3.20)*	25.53 (3.14)	25.63 (3.19)
Systolic blood pressure, mean (SD)	132.01 (8.15)	132.50 (8.81)*	131.62 (8.06)	132.89 (8.86)†	131.62 (8.13)	132.89 (8.79)†	131.49 (8.07)	133.02 (8.82)†	131.62 (8.06)	132.89 (8.85)†
Diastolic blood pressure, mean (SD)	78.75 (5.80)	77.15 (6.32)†	78.03 (5.91)	77.87 (6.32)	78.05 (5.87)	77.86 (6.35)	77.95 (5.89)	77.95 (6.34)	78.05 (5.91)	77.85 (6.32)
Current smoking, n (%)	618 (16.1)	611 (15.9)	615 (16.0)	614 (16.0)	607 (15.9)	622 (16.1)	617 (16.1)	612 (15.9)	612 (15.9)	617 (16.1)
Medication history, n (%)										
Diabetes	708 (18.4)	741 (19.3)	714 (18.6)	735 (19.2)	694 (18.2)	755 (19.6)	736 (19.2)	713 (18.6)	714 (18.6)	735 (19.1)
Dyslipidemia	1330 (34.6)	1468 (38.3)†	1335 (34.7)	1463 (38.2)‡	1311 (34.3)	1487 (38.6)†	1331 (34.7)	1467 (38.2)‡	1334 (34.8)	1464 (38.1)†
CVD history	654 (17.0)	682 (17.8)	653 (17.0)	683 (17.8)	647 (16.9)	689 (17.9)	652 (17.0)	684 (17.8)	651 (17.0)	685 (17.8)
Chronic kidney dysfunction	44 (1.1)	58 (1.5)	45 (1.2)	57 (1.5)	41 (1.1)	61 (1.6)	41 (1.1)	61 (1.6)*	45 (1.2)	57 (1.5)
eGFR, mean (SD)	108.57 (22.76)	108.83 (23.41)	108.65 (22.67)	108.75 (23.50)	108.54 (22.56)	108.86 (23.60)	109.09 (22.79)	108.31 (23.38)	108.67 (22.67)	108.74 (23.50)
Antihypertensive medications, n (%)										
ARB	2634 (68.5)	2672 (69.7)	2627 (68.3)	2679 (69.9)	2619 (68.5)	2687 (69.7)	2638 (68.8)	2668 (69.4)	2621 (68.3)	2685 (69.9)
CCB	3047 (79.2)	3030 (79.1)	3037 (79.0)	3040 (79.3)	3034 (79.4)	3043 (78.9)	3048 (79.5)	3029 (78.8)	3031 (79.0)	3046 (79.3)
Diuretic	306 (8.0)	374 (9.8)‡	305 (7.9)	375 (9.8)‡	304 (8.0)	376 (9.8)‡	298 (7.8)	382 (9.9)†	304 (7.9)	376 (9.8)‡
β-blockers	124 (3.2)	147 (3.8)	126 (3.3)	145 (3.8)	118 (3.1)	153 (4.0)*	117 (3.1)	154 (4.0)*	126 (3.3)	145 (3.8)
No. of antihypertensive medication classes										
0	93 (2.4)	131 (3.4)†	95 (2.5)	129 (3.4)†	92 (2.4)	132 (3.4)†	90 (2.3)	134 (3.5)†	95 (2.5)	129 (3.4)†
1	1699 (44.2)	1554 (40.6)†	1714 (44.6)	1539 (40.2)†	1692 (44.3)	1561 (40.5)†	1687 (44.0)	1566 (40.8)†	1710 (44.6)	1543 (40.2)†
2	1737 (45.2)	1767 (46.1)†	1715 (44.6)	1789 (46.7)†	1722 (45.0)	1782 (46.2)†	1749 (45.6)	1755 (45.7)†	1712 (44.6)	1792 (46.7)†
≥3	317 (8.2)	380 (9.9)†	321 (8.3)	376 (9.8)†	317 (8.3)	380 (9.9)†	310 (8.1)	387 (10.1)†	320 (8.3)	377 (9.8)†
Other medications										
Aspirin	260 (6.8)	391 (10.2)†	280 (7.3)	371 (9.7)†	276 (7.2)	375 (9.7)†	276 (7.2)	375 (9.8)†	279 (7.3)	372 (9.7)†
Statin	650 (16.9)	808 (21.1)†	668 (17.4)	790 (20.6)†	655 (17.1)	803 (20.8)†	674 (17.6)	784 (20.4)‡	667 (17.4)	791 (20.6)†
Antidiabetic agent	596 (15.5)	660 (17.2)*	604 (15.7)	652 (17.0)	593 (15.5)	663 (17.2)*	620 (16.2)	636 (16.6)	604 (15.7)	652 (17.0)
Framingham score ≥15	2985 (77.6)	3011 (78.6)	2963 (77.1)	3033 (79.1)*	2944 (77.0)	3052 (79.2)*	2984 (77.8)	3012 (78.4)	2957 (77.1)	3039 (79.1)*

BMI is the weight in kilograms divided by the square of the height in meters. Chronic kidney dysfunction was defined as an estimated glomerular filtration rate of <60 mL per minute per 1.73 m^2. eGFR was calculated according to the 4-component Modification of Diet in Renal Disease study prediction equation. A Framingham Risk Score of 15% or higher indicates a high 10-year risk of cardiovascular disease. ARB indicates angiotensin receptor blocker; ARV, average real variability; BMI, body mass index; CCB, calcium channel blocker; CV, coefficient of variation; CVD, cardiovascular disease; eGFR, glomerular filtration rate estimated from the serum creatinine; and VIM, variability independent of the mean.

*$P \le 0.05$.

†$P \le 0.001$.

‡$P \le 0.01$.

sensitivity analysis, the association between diastolic BPV and outcome events including the primary end point, MACE, and ACS in the standard treatment group were proven in the sensitivity analysis (adjusted HR, ≥1.28; $P \le 0.019$; Tables S5 and S7), and there was no evidence that a greater systolic BPV was related to a higher risk of adverse outcomes ($P \ge 0.095$; Table S5). Furthermore, the result that neither the systolic BPV nor the diastolic BPV had prognostic significance for the primary outcome, MACE, and ACS in the intensive treatment group was not changed ($P \ge 0.11$; Tables S4 and S8).

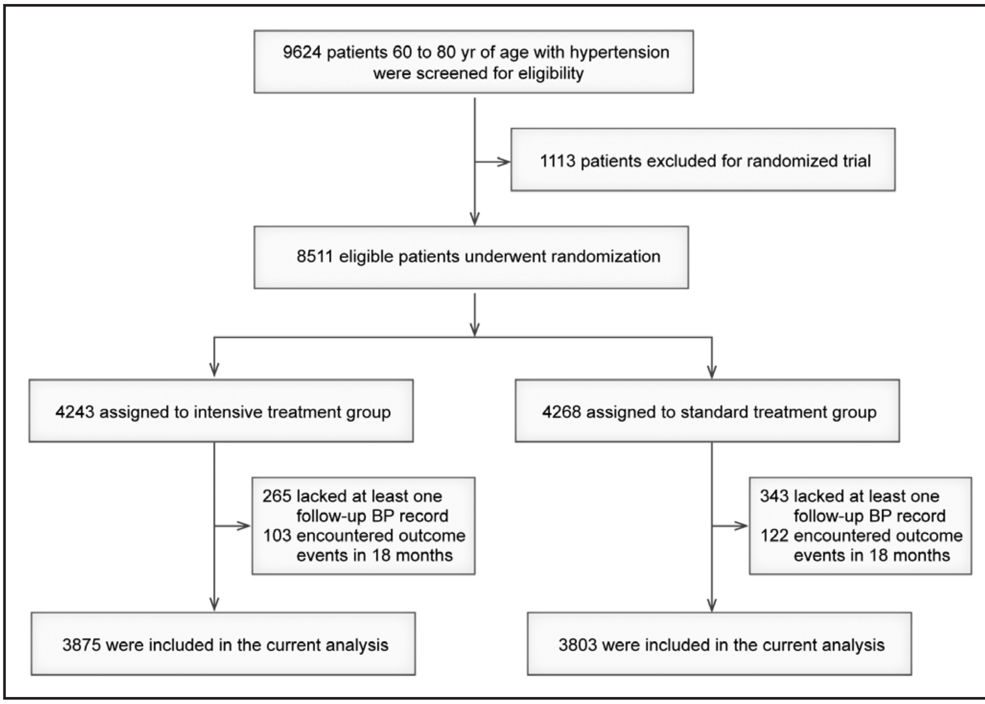

Figure 1. Flowchart of participant selection.
BP indicates blood pressure.

Factors Correlated With BPV

Multivariable-adjusted regression models revealed that intake of beta-blockers was associated with greater diastolic BPV (using the coefficient of variation for the analysis), while the standard antihypertension treatment and the use of aspirin were related to a lower diastolic BPV (Figure 2).

A greater systolic BPV was significantly associated with a history of hyperlipidemia, Framingham score, and intake of diuretics and beta-blockers, while a lower systolic BPV was associated with the male sex and receiving standard BP control (Figure 3).

DISCUSSION

In this study, we investigated the associations between adverse clinical outcomes and visit-to-visit BPV among 7678 patients enrolled in the STEP trial. The main findings of our research were (1) diastolic BPV independently predicted worse clinical outcomes, including the primary outcome, MACE, and ACS in older adults with hypertension in the standard treatment group, whereas there was no observed association between systolic BPV and cardiovascular events; (2) the association between diastolic BPV and cardiovascular events was no longer significant among patients receiving intensive BP control; (3) sensitivity analyses using more BP records during the follow-up period to calculate the BPV confirmed the abovementioned results.

Interestingly, the prognostic value of diastolic variability was the most prominent among the standard treatment group in our study. Our finding was consistent with the post hoc analysis of the Blood Pressure and Clinical Outcome in TIA and IS trial (BOSS), in which cardiovascular events were only related to diastolic BPV (defined as the SD and coefficient of variation of the DBP) rather than systolic BPV in patients with a history of ischemic stroke and transient ischemic attack.[22] Similar findings were also reported in the Action to Control Cardiovascular Risk in Diabetes trial (ACCORD) in which diastolic BPV was strongly associated with coronary heart disease, especially in patients with a low baseline BP.[23] In addition, diastolic BPV has been reported as a risk factor for cognitive decline and stroke recurrence.[9,22,24] These findings might indicate that diastolic BPV becomes more important for older adults with hypertension and that it is essential to monitor the DBP in older adults. The underlying mechanism for the significance of diastolic BPV in older adults might be the fluctuation and reduction of coronary perfusion during diastole, which leads to coronary hypoperfusion and exacerbates myocardial ischemia.[25,26] It is well known that the DBP plays an essential role in maintaining coronary perfusion. Thus, a greater diastolic BPV might represent long-term unstable blood supply to the myocardium. The potential adverse effects owing to excessive BP reduction might also play a role. A recent analysis of the Systolic Blood Pressure Intervention Trial (SPRINT) suggested that greater diastolic BPV was specifically linked to potential hypoperfusion-related adverse events.[25] Furthermore, the increased arterial stiffness, reduction in aortic distensibility, and impaired autoregulation ability in older adults might increase the risks of target

Table 2. Associations of BPV With Primary End Point, MACE, and ACS in the Standard Treatment Group

	Primary		MACE		ACS	
Variability model	HR (95% CI)	*P* value	HR (95% CI)	*P* value	HR (95% CI)	*P* value
Systolic BPV						
Unadjusted models						
CV	1.14 (0.96–1.36)	0.130	1.15 (0.95–1.38)	0.156	1.15 (0.93–1.43)	0.192
SD	1.16 (0.98–1.37)	0.088	1.15 (0.96–1.39)	0.136	1.16 (0.94–1.43)	0.172
Delta	1.18 (1.00–1.39)	0.056	1.16 (0.96–1.40)	0.119	1.17 (0.94–1.44)	0.153
ARV	1.09 (0.92–1.30)	0.316	1.08 (0.89–1.31)	0.458	1.04 (0.83–1.30)	0.708
VIM	1.14 (0.96–1.36)	0.121	1.15 (0.95–1.38)	0.152	1.15 (0.93–1.43)	0.188
Adjusted models						
CV	1.15 (0.96–1.37)	0.127	1.16 (0.96–1.40)	0.128	1.18 (0.95–1.46)	0.135
SD	1.15 (0.96–1.37)	0.129	1.16 (0.95–1.40)	0.137	1.17 (0.95–1.46)	0.148
Delta	1.16 (0.98–1.38)	0.091	1.16 (0.96–1.41)	0.125	1.17 (0.95–1.46)	0.141
ARV	1.07 (0.89–1.28)	0.483	1.07 (0.87–1.30)	0.528	1.04 (0.82–1.31)	0.753
VIM	1.15 (0.96–1.37)	0.127	1.16 (0.96–1.40)	0.130	1.18 (0.95–1.46)	0.138
Diastolic BPV						
Unadjusted models						
CV	1.29 (1.10–1.51)	0.002	1.27 (1.07–1.52)	0.007	1.33 (1.09–1.61)	0.004
SD	1.30 (1.11–1.52)	0.001	1.27 (1.07–1.52)	0.007	1.33 (1.10–1.61)	0.003
Delta	1.33 (1.14–1.56)	<0.001	1.29 (1.08–1.54)	0.004	1.36 (1.13–1.65)	0.002
ARV	1.22 (1.03–1.44)	0.018	1.21 (1.01–1.45)	0.043	1.21 (0.98–1.49)	0.072
VIM	1.30 (1.11–1.52)	0.001	1.27 (1.07–1.52)	0.007	1.33 (1.10–1.61)	0.003
Adjusted models						
CV	1.24 (1.06–1.47)	0.009	1.21 (1.01–1.46)	0.040	1.28 (1.05–1.57)	0.017
SD	1.26 (1.07–1.48)	0.005	1.23 (1.03–1.48)	0.026	1.30 (1.06–1.58)	0.010
Delta	1.29 (1.10–1.52)	0.002	1.25 (1.04–1.51)	0.016	1.33 (1.09–1.62)	0.005
ARV	1.21 (1.02–1.43)	0.029	1.20 (0.99–1.45)	0.059	1.22 (0.98–1.51)	0.076
VIM	1.26 (1.07–1.47)	0.005	1.23 (1.02–1.48)	0.026	1.30 (1.06–1.58)	0.011

HRs (95% CI) express the difference in primary end point, MACE, and ACS associated with 1-SD increment in visit-to-visit variability indexes in the standard treatment group. Adjusted models accounted for sex, age, body mass index, baseline CKD status, diabetes, hyperlipidemia, cardiovascular disease, current smoking, mean corresponding blood pressure during the follow-up period, the number of antihypertensive medication classes received at the 6-mo study visit, use of antihypertensive medications by drug class, that is, diuretics, β-blockers, inhibitors of the renin-angiotensin, calcium channel blockers at 6-mo, baseline intake of aspirin, lipid-lowering drugs, and antidiabetic agents. ACS indicates acute coronary syndrome; ARV average real variability; BPV, blood pressure variability; CKD, chronic kidney disease; CV, coefficient of variation; HR, hazard ratio; MACE, major adverse cardiac events; and VIM, variability independent of the mean.

organ hypoperfusion. In the present study, the association between diastolic BPV and cardiovascular risk was no longer significant among patients receiving intensive BP control, which might indicate that controlling the BP to a low level (SBP <130 mm Hg in the STEP trial) eliminated the prognostic value of the BPV for cardiovascular events. One potential reason for this is the lower BP achieved by intensive treatment (SBP of 126.7 mm Hg and DBP of 76.4 mm Hg during follow-up). Another reason might be that the relatively low incidence of cardiovascular events in the intensive treatment group led to a limited statistical power to detect significant associations.

Regarding the association between systolic BPV and cardiovascular disease, our analysis has yielded results that are inconsistent with previous research, such as the Anglo-Scandinavian Cardiac Outcomes Trial-Blood Pressure Lowering Arm trial (ASCOT-BPLA) and the Stabilization of Atherosclerotic Plaque by Initiation of Darapladib Therapy trial (STABILITY).[8,26] Rothwell et al[8] reported that visit-to-visit systolic BPV (defined as the SD and variability independent of the mean of the systolic BP) is a powerful predictor of stroke and coronary events in patients with treated hypertension enrolled in the ASCOT-BPLA trial. The STABILITY investigators demonstrated that both the highest tertile of systolic BPV (defined as the SD of the systolic BP and diastolic BP) and highest tertile of diastolic BPV were associated with the primary end point (95% CIs, 1.10–1.53 and 1.18–1.62, respectively). However, the lack of association between systolic BPV and cardiovascular events has also been reported in the

Table 3. Associations of BPV With Primary End Point, MACE, and ACS in the Intensive Treatment Group

Variability model	Primary		MACE		ACS	
	HR (95% CI)	*P* value	HR (95% CI)	*P* value	HR (95% CI)	*P* value
Systolic BPV						
Unadjusted models						
CV	1.08 (0.89–1.31)	0.446	1.05 (0.85–1.30)	0.654	1.00 (0.79–1.27)	0.995
SD	1.09 (0.90–1.32)	0.373	1.05 (0.85–1.31)	0.627	1.00 (0.79–1.27)	0.976
Delta	1.11 (0.91–1.34)	0.298	1.06 (0.86–1.31)	0.584	1.01 (0.80–1.28)	0.930
ARV	1.10 (0.91–1.33)	0.311	1.04 (0.84–1.29)	0.724	0.94 (0.73–1.21)	0.628
VIM	1.08 (0.89–1.31)	0.432	1.05 (0.85–1.30)	0.648	1.00 (0.79–1.27)	0.991
Adjusted models						
CV	1.03 (0.83–1.27)	0.810	1.03 (0.81–1.31)	0.811	0.98 (0.75–1.28)	0.901
SD	1.03 (0.82–1.28)	0.827	1.03 (0.80–1.31)	0.839	0.98 (0.74–1.29)	0.879
Delta	1.04 (0.84–1.30)	0.714	1.03 (0.81–1.32)	0.802	0.99 (0.75–1.30)	0.922
ARV	1.02 (0.83–1.27)	0.835	0.99 (0.78–1.26)	0.917	0.88 (0.67–1.17)	0.384
VIM	1.03 (0.83–1.27)	0.813	1.03 (0.81–1.31)	0.816	0.98 (0.75–1.28)	0.897
Diastolic BPV						
Unadjusted models						
CV	1.15 (0.95–1.40)	0.143	1.20 (0.97–1.47)	0.087	1.18 (0.94–1.48)	0.146
SD	1.13 (0.93–1.37)	0.204	1.17 (0.95–1.44)	0.148	1.15 (0.92–1.44)	0.217
Delta	1.14 (0.94–1.38)	0.184	1.16 (0.94–1.43)	0.156	1.15 (0.91–1.44)	0.240
ARV	1.03 (0.84–1.25)	0.800	1.01 (0.81–1.26)	0.946	1.00 (0.78–1.27)	0.975
VIM	1.13 (0.94–1.37)	0.202	1.17 (0.95–1.44)	0.146	1.15 (0.92–1.45)	0.214
Adjusted models						
CV	1.10 (0.91–1.33)	0.335	1.13 (0.91–1.39)	0.269	1.12 (0.89–1.40)	0.345
SD	1.10 (0.90–1.33)	0.343	1.13 (0.92–1.40)	0.252	1.12 (0.89–1.40)	0.349
Delta	1.11 (0.91–1.34)	0.307	1.13 (0.91–1.40)	0.260	1.11 (0.88–1.40)	0.363
ARV	1.00 (0.82–1.23)	0.963	0.98 (0.78–1.23)	0.858	0.97 (0.76–1.23)	0.796
VIM	1.10 (0.91–1.33)	0.343	1.13 (0.92–1.40)	0.253	1.12 (0.89–1.40)	0.349

HRs (95% CI) express the difference in primary end point, MACE, and ACS associated with 1-SD increment in visit-to-visit variability indexes in the intensive treatment group. Adjusted models accounted for sex, age, body mass index, baseline CKD status, diabetes, hyperlipidemia, cardiovascular disease, current smoking, mean corresponding blood pressure during the follow-up period, the number of antihypertensive medication classes received at the 6-mo study visit, use of antihypertensive medications by drug class, that is, diuretics, β-blockers, inhibitors of the renin-angiotensin, calcium channel blockers at 6-mo, baseline intake of aspirin, lipid-lowering drugs, and antidiabetic agents. ACS indicates acute coronary syndrome; ARV average real variability; BPV, blood pressure variability; CKD, chronic kidney disease; CV, coefficient of variation; HR, hazard ratio; MACE, major adverse cardiac events; and VIM, variability independent of the mean.

reanalysis of the SPRINT trial and the European Lacidipine Study on Atherosclerosis trial (ELSA), which was consistent with our results.[12,27] There are some potential reasons for this interstudy discrepancy regarding the prognostic value of the systolic BPV. First, the STEP trial was a well-controlled cohort, patients enrolled in the STEP trial all received standard antihypertensive therapy and regular BP screenings during follow-up; thus, this cohort achieved a low mean BP compared with other cohorts. Second, as the patients received regular office BP monitoring as well as home BP monitoring, the patients in our present study had a relatively stable systolic BP. In other words, they had a lower systolic BPV compared with previous studies (8.2–10.0 in the ASCOT-BPLA trial and 9.8 in the STABILITY trial). Third, differences between studies in factors such as the study design, population selection, and comorbidities, BP measurement approach, time interval between visits used for calculating BPV, and BPV indexes may also have contributed to the conflicting results. The inconsistent BP measurement approaches and calculated indexes between different studies have made it difficult to clarify the prognostic significance of BPV. Further study is warranted to clarify the definition of BPV and the optimal approach for measuring visit-to-visit BPV.

Visit-to-visit BPV is reportedly affected by environmental factors (season and temperature changes), biological factors (older age, women), comorbidities (previous cardiovascular disease or chronic kidney disease), and medication factors (medication adherence, intake of calcium channel blockers and angiotensin receptor

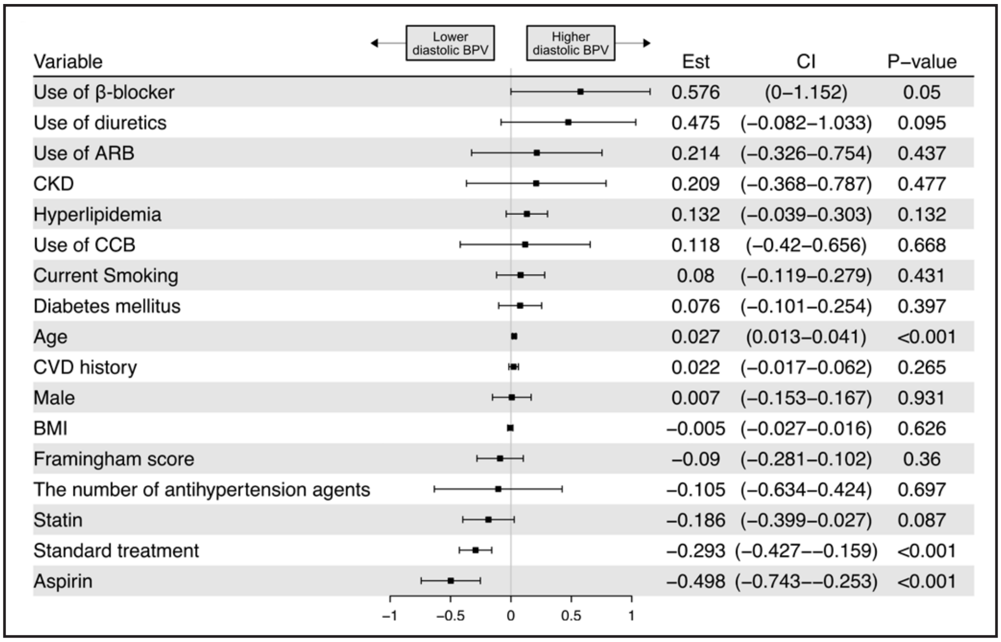

Figure 2. Determinants of visit-to-visit diastolic blood pressure variability (BPV) from multivariable-adjusted linear regression models.
Using coefficient of variation (CV) to represent BPV. ARB indicates angiotensin receptor blocker; BMI, body mass index; CCB, calcium channel blocker; CKD, chronic kidney disease; and CVD, cardiovascular disease.

blockers).[12,28–30] In the STEP cohort, we found that a higher FRS was associated with greater systolic BPV. The underlying mechanism might be the strong moderation effect of arterial remodeling in the relationship between BPV and the FRS. A higher FRS is reportedly associated with more severe arterial stiffness.[31] Furthermore, the remodeling and atherosclerosis of arteries, including the aorta and carotid, may also be the pathophysiology of the unstable BP and greater BPV.[19,32] Interestingly, the intake of aspirin was associated with lowing BPV, the mechanisms responsible for its beneficial influence might be the effects of aspirin on

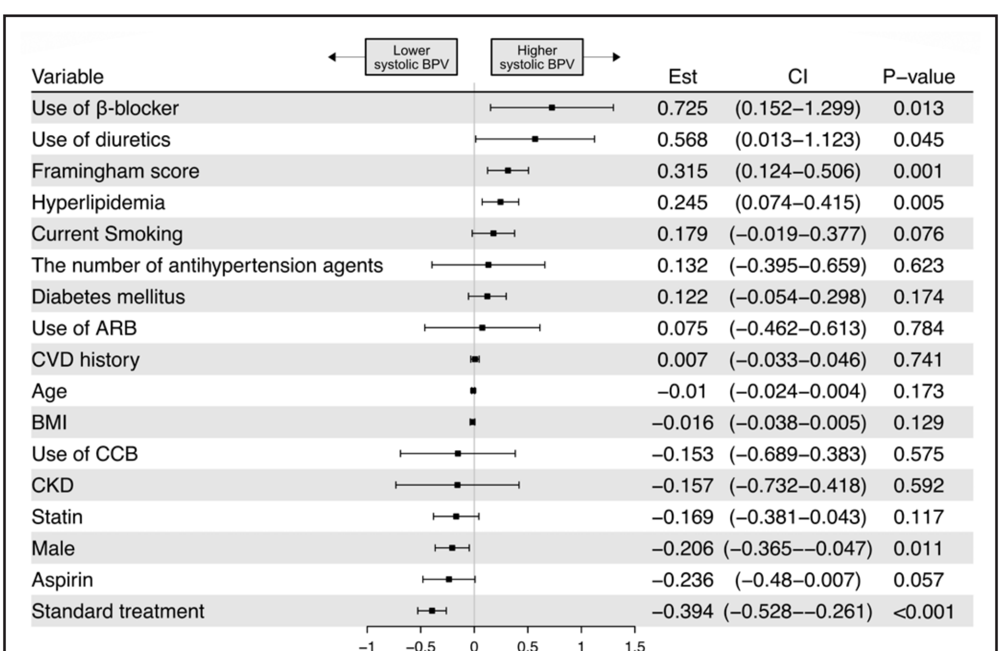

Figure 3. Determinants of visit-to-visit systolic blood pressure variability (BPV) from multivariable-adjusted linear regression models.
Using coefficient of variation (CV) to represent BPV. ARB indicates angiotensin receptor blocker; BMI, body mass index; CCB, calcium channel blocker; CKD, chronic kidney disease; and CVD, cardiovascular disease.

reducing atherosclerosis and further improving arterial stiffness.[33–36] Antihypertensive medication can also play an important role in the development of BPV. Our study found that the patients who received intensive BP control had a greater BPV, and the regression models showed that the treatment strategy was an independent factor influencing the BPV. The probable reason for this may be that patients who received intensive treatment had larger absolute BP reductions and a higher incidence of hypotension.[37] In addition, patients in the intensive treatment group received more antihypertensive agent titration to achieve the lower SBP target. As for medication class which might influence BPV, we found β-blocker was related a higher BPV in this cohort, which is consistent with previous studies such as ASCOT-BPLA and Medical Research Council trial (MRC).[38] The vasoconstriction effect of β-blocker such as atenolol can lead to reduction of arterial compliance and peripheral vascular resistance was perhaps the most likely explanation. And the reduced effectiveness of β-blocker on arterial stiffness might also make contributions.[39,40] Given that the antihypertensive drug was not the study arm of the STEP trial, further study is needed to be done to clarify the effect of different antihypertensive drugs on BPV.

The current study had the following strengths. (1) Our research was able to examine the value of BPV in different treatment strategy groups owing to the randomized design of the STEP trial. (2) We investigated the association between cardiovascular events and visit-to-visit BPV as measured by 5 different indexes, which avoid the bias caused by the analysis of a single index. (3) The STEP trial was a well-organized multicenter trial with a large sample size and carefully recorded data, which increases the reliability of our study results. Our analysis also has several limitations. First, our study was a post hoc observation analysis of the STEP trial. However, the assessment of the BPV was not the direct purpose of the STEP trial. Therefore, residual confounding factors and bias may influence the present results. Second, as the median follow-up period was only 3.34 years, the numbers of some clinical outcomes (such as cardiovascular death, stroke, and heart failure) were relatively few, which limits our ability to evaluate the association between BPV and these clinical events. Third, the STEP trial enrolled older adults with hypertension, which is a very select population. This may limit the generalizability of our findings.

Perspectives

Diastolic BPV was associated with clinical adverse events in older adults with hypertension but not in patients who received intensive BP control. Our results highlight the potential role of monitoring diastolic BPV in older adults and the possible need to monitor the diastolic BP in clinical practice. Future investigation in large, population-based studies is warranted to build a strong evidence-based understanding of the predictive power of BPV indexes for cardiovascular events when targeting a lower BP.

ARTICLE INFORMATION

Received October 21, 2022; accepted March 22, 2023.

Affiliations

Hypertension Center, Fuwai Hospital, State Key Laboratory of Cardiovascular Disease of China, National Center for Cardiovascular Diseases of China, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China (J.Y., Q.S., J.J.B., J.C.). Department of Cardiology, Kailuan General Hospital, Tangshan, Hebei, China (S.W.). Department of Cardiology, Qilu Hospital of Shandong University, Jinan, China (P.B.). Department of Cardiology, Beijing Pinggu Hospital, China (Y.F.L.).

Acknowledgments

The authors thank all the staff members of the STEP (Strategy of Blood Pressure Intervention in the Elderly Hypertensive Patients) Team for their contributions. The authors thank Kelly Zammit, BVSc, from Liwen Bianji (Edanz), for editing the English text of a draft of this article.

Sources of Funding

This work was supported by the National Natural Science Foundation of China (Project ID. 81825002).

Disclosures

None.

REFERENCES

1. Parati G, Ochoa JE, Lombardi C, Bilo G. Assessment and management of blood-pressure variability. *Nat Rev Cardiol.* 2013;10:143–155. doi: 10.1038/nrcardio.2013.1
2. Parati G, Ochoa JE, Lombardi C, Bilo G. Blood pressure variability: assessment, predictive value, and potential as a therapeutic target. *Curr Hypertens Rep.* 2015;17:537. doi: 10.1007/s11906-015-0537-1
3. Lau KK, Wong YK, Chan YH, Teo KC, Chan KH, Wai Li LS, Cheung RTF, Siu CW, Ho SL, Tse HF. Visit-to-visit blood pressure variability as a prognostic marker in patients with cardiovascular and cerebrovascular diseases--relationships and comparisons with vascular markers of atherosclerosis. *Atherosclerosis.* 2014;235:230–235. doi: 10.1016/j.atherosclerosis.2014.04.015
4. Muntner P, Whittle J, Lynch AI, Colantonio LD, Simpson LM, Einhorn PT, Levitan EB, Whelton PK, Cushman WC, Louis GT, et al. Visit-to-visit variability of blood pressure and coronary heart disease, stroke, heart failure, and mortality: a cohort study. *Ann Intern Med.* 2015;163:329–338. doi: 10.7326/M14-2803
5. Hata J, Arima H, Rothwell PM, Woodward M, Zoungas S, Anderson C, Patel A, Neal B, Glasziou P, Hamet P, et al; ADVANCE Collaborative Group. Effects of visit-to-visit variability in systolic blood pressure on macrovascular and microvascular complications in patients with type 2 diabetes mellitus: the ADVANCE trial. *Circulation.* 2013;128:1325–1334. doi: 10.1161/CIRCULATIONAHA.113.002717
6. Wei FF, Zhou Y, Thijs L, Xue R, Dong B, He X, Liang W, Wu Y, Jiang J, Tan W, et al. Visit-to-visit blood pressure variability and clinical outcomes in patients with heart failure with preserved ejection fraction. *Hypertension.* 2021;77:1549–1558. doi: 10.1161/HYPERTENSIONAHA.120.16757
7. Clark D 3rd, Nicholls SJ, St John J, Elshazly MB, Ahmed HM, Khraishah H, Nissen SE, Puri R. Visit-to-visit blood pressure variability, coronary atheroma progression, and clinical outcomes. *JAMA Cardiol.* 2019;4:437–443. doi: 10.1001/jamacardio.2019.0751
8. Rothwell PM, Howard SC, Dolan E, O'Brien E, Dobson JE, Dahlöf B, Sever PS, Poulter NR. Prognostic significance of visit-to-visit variability, maximum systolic blood pressure, and episodic hypertension. *Lancet.* 2010;375:895–905. doi: 10.1016/S0140-6736(10)60308-X
9. de Havenon A, Fino NF, Johnson B, Wong KH, Majersik JJ, Tirschwell D, Rost N. Blood pressure variability and cardiovascular outcomes in patients with prior stroke: a secondary analysis of PRoFESS. *Stroke.* 2019;50:3170–3176. doi: 10.1161/STROKEAHA.119.026293
10. Arashi H, Ogawa H, Yamaguchi J, Kawada-Watanabe E, Hagiwara N. Impact of visit-to-visit variability and systolic blood pressure control on subsequent

outcomes in hypertensive patients with coronary artery disease (from the HIJ-CREATE substudy). *Am J Cardiol*. 2015;116:236–242. doi: 10.1016/j.amjcard.2015.04.011
11. Schutte R, Thijs L, Liu YP, Asayama K, Jin Y, Odili A, Gu Y-M, Kuznetsova T, Jacobs L, Staessen JA. Within-subject blood pressure level--not variability--predicts fatal and nonfatal outcomes in a general population. *Hypertension*. 2012;60:1138–1147. doi: 10.1161/HYPERTENSIONAHA.112.202143
12. Chang TI, Reboussin DM, Chertow GM, Cheung AK, Cushman WC, Kostis WJ, Parati G, Raj D, Riessen E, Shapiro B, et al; SPRINT Research Group*. Visit-to-visit office blood pressure variability and cardiovascular outcomes in SPRINT (Systolic Blood Pressure Intervention Trial). *Hypertension*. 2017;70:751–758. doi: 10.1161/HYPERTENSIONAHA.117.09788
13. Zhang W, Zhang S, Deng Y, Wu S, Ren J, Sun G, Yang J, Jiang Y, Xu X, Wang T-D, et al; STEP Study Group. Trial of intensive blood-pressure control in older patients with hypertension. *N Engl J Med*. 2021;385:1268–1279. doi: 10.1056/NEJMoa2111437
14. Wright JT Jr, Williamson JD, Whelton PK, Snyder JK, Sink KM, Rocco MV, Reboussin DM, Rahman M, Oparil S, Lewis CE, et al; Group SR. A randomized trial of intensive versus standard blood-pressure control. *N Engl J Med*. 2015;373:2103–2116. doi: 10.1056/NEJMoa1511939
15. Xie X, Atkins E, Lv J, Bennett A, Neal B, Ninomiya T, Woodward M, MacMahon S, Turnbull F, Hillis GS, et al. Effects of intensive blood pressure lowering on cardiovascular and renal outcomes: updated systematic review and meta-analysis. *Lancet*. 2016;387:435–443. doi: 10.1016/S0140-6736(15)00805-3
16. Zhang S, Wu S, Ren J, Chen X, Zhang X, Feng Y, Zhou X, Zhu B, Yang J, Tian G, et al; STEP Study Group. Strategy of Blood Pressure Intervention in the Elderly Hypertensive Patients (STEP): rational, design, and baseline characteristics for the main trial. *Contemp Clin Trials*. 2020;89:105913. doi: 10.1016/j.cct.2019.105913
17. Tully PJ, Dartigues JF, Debette S, Helmer C, Artero S, Tzourio C. Dementia risk with antihypertensive use and blood pressure variability: a cohort study. *Neurology*. 2016;87:601–608. doi: 10.1212/WNL.0000000000002946
18. Mena L, Pintos S, Queipo NV, Aizpurua JA, Maestre G, Sulbaran T. A reliable index for the prognostic significance of blood pressure variability. *J Hypertens*. 2005;23:505–511. doi: 10.1097/01.hjh.0000160205.81652.5a
19. Nagai M, Hoshide S, Ishikawa J, Shimada K, Kario K. Visit-to-visit blood pressure variations: new independent determinants for carotid artery measures in the elderly at high risk of cardiovascular disease. *J Am Soc Hypertens*. 2011;5:184–192. doi: 10.1016/j.jash.2011.03.001
20. Wilson PW, D'Agostino RB, Levy D, Belanger AM, Silbershatz H, Kannel WB. Prediction of coronary heart disease using risk factor categories. *Circulation*. 1998;97:1837–1847. doi: 10.1161/01.cir.97.18.1837
21. Levey AS, Coresh J, Greene T, Stevens LA, Zhang YL, Hendriksen S, Kusek JW, Van Lente F; Chronic Kidney Disease Epidemiology Collaboration. Using standardized serum creatinine values in the modification of diet in renal disease study equation for estimating glomerular filtration rate. *Ann Intern Med*. 2006;145:247–254. doi: 10.7326/0003-4819-145-4-200608150-00004
22. Dai L, Cheng A, Hao X, Xu J, Zuo Y, Wang A, Meng X, Li H, Wang Y, Zhao X, et al. Different contribution of SBP and DBP variability to vascular events in patients with stroke. *Stroke Vasc Neurol*. 2020;5:110–115. doi: 10.1136/svn-2019-000278
23. Nuyujukian DS, Zhou JJ, Koska J, Reaven PD. Refining determinants of associations of visit-to-visit blood pressure variability with cardiovascular risk: results from the Action to Control Cardiovascular Risk in Diabetes Trial. *J Hypertens*. 2021;39:2173–2182. doi: 10.1097/HJH.0000000000002931
24. Peters R, Xu Y, Eramudugolla R, Sachdev PS, Cherbuin N, Tully PJ, Mortby ME, Anstey KJ. Diastolic blood pressure variability in later life may be a key risk marker for cognitive decline. *Hypertension*. 2022;79:1037–1044. doi: 10.1161/HYPERTENSIONAHA.121.18799
25. Mezue K, Goyal A, Pressman GS, Matthew R, Horrow JC, Rangaswami J. Blood pressure variability predicts adverse events and cardiovascular outcomes in SPRINT. *J Clin Hypertens (Greenwich)*. 2018;20:1247–1252. doi: 10.1111/jch.13346
26. Vidal-Petiot E, Stebbins A, Chiswell K, Ardissino D, Aylward PE, Cannon CP, Ramos Corrales MA, Held C, López-Sendón JL, Stewart RAH, et al; STABILITY Investigators. Visit-to-visit variability of blood pressure and cardiovascular outcomes in patients with stable coronary heart disease. Insights from the STABILITY trial. *Eur Heart J*. 2017;38:2813–2822. doi: 10.1093/eurheartj/ehx250
27. Mancia G, Facchetti R, Parati G, Zanchetti A. Visit-to-visit blood pressure variability, carotid atherosclerosis, and cardiovascular events in the European Lacidipine Study on Atherosclerosis. *Circulation*. 2012;126:569–578. doi: 10.1161/CIRCULATIONAHA.112.107565
28. Schutte AE, Kollias A, Stergiou GS. Blood pressure and its variability: classic and novel measurement techniques. *Nat Rev Cardiol*. 2022;19:643–654. doi: 10.1038/s41569-022-00690-0
29. Guthmann R, Davis N, Brown M, Elizondo J. Visit frequency and hypertension. *J Clin Hypertens (Greenwich)*. 2005;7:327–332. doi: 10.1111/j.1524-6175.2005.04371.x
30. King CC, Bartels CM, Magnan EM, Fink JT, Smith MA, Johnson HM. The importance of frequent return visits and hypertension control among US young adults: a multidisciplinary group practice observational study. *J Clin Hypertens (Greenwich)*. 2017;19:1288–1297. doi: 10.1111/jch.13096
31. Jin L, Tong L, Shen C, Du L, Mao J, Liu L, Li Z. Association of arterial stiffness indices with framingham cardiovascular disease risk score. *RCM*. 2022;23:287. doi: 10.31083/j.rcm2308287
32. Miyauchi S, Nagai M, Dote K, Kato M, Oda N, Kunita E, Kagawa E, Yamane A, Higashihara T, Takeuchi A, et al. Visit-to-visit blood pressure variability and arterial stiffness: which came first: the chicken or the egg? *Curr Pharm Des*. 2019;25:685–692. doi: 10.2174/1381612825666190329122024
33. Smith JB, Araki H, Lefer AM. Thromboxane A2, prostacyclin and aspirin: effects on vascular tone and platelet aggregation. *Circulation*. 1980;62:V19–V25.
34. Raymenants E, Yang B, Nicolini F, Behrens P, Lawson D, Mehta JL. Verapamil and aspirin modulate platelet-mediated vasomotion in arterial segments with intact or disrupted endothelium. *J Am Coll Cardiol*. 1993;22:684–689. doi: 10.1016/0735-1097(93)90177-3
35. Pietri P, Vlachopoulos C, Terentes-Printzios D, Xaplanteris P, Aznaouridis K, Petrocheilou K, Stefanadis C. Beneficial effects of low-dose aspirin on aortic stiffness in hypertensive patients. *Vasc Med*. 2014;19:452–457. doi: 10.1177/1358863X14556695
36. Yasumasu T, Takahara K, Otsuji Y. Low-dose aspirin inhibits cardiac sympathetic activation and vagal withdrawal response to morning rising. *J Cardiovasc Pharmacol*. 2017;70:239–244. doi: 10.1097/FJC.0000000000000509
37. Smith TO, Sillito JA, Goh CH, Abdel-Fattah A-R, Einarsson A, Soiza RL, Mamas MA, Tan MP, Potter JF, Loke YK, et al. Association between different methods of assessing blood pressure variability and incident cardiovascular disease, cardiovascular mortality and all-cause mortality: a systematic review. *Age Ageing*. 2020;49:184–192. doi: 10.1093/ageing/afz178
38. Rothwell PM, Howard SC, Dolan E, O'Brien E, Dobson JE, Dahlöf B, Poulter NR, Sever PS; ASCOT-BPLA and MRC Trial Investigators. Effects of beta blockers and calcium-channel blockers on within-individual variability in blood pressure and risk of stroke. *Lancet Neurol*. 2010;9:469–480. doi: 10.1016/S1474-4422(10)70066-1
39. Dudenbostel T, Glasser SP. Effects of antihypertensive drugs on arterial stiffness. *Cardiol Rev*. 2012;20:259–263. doi: 10.1097/CRD.0b013e31825d0a44
40. Rothwell PM. Limitations of the usual blood-pressure hypothesis and importance of variability, instability, and episodic hypertension. *Lancet*. 2010;375:938–948. doi: 10.1016/S0140-6736(10)60309-1